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AI-Powered Personalized Financial Advisor: Empowering Smarter Financial Decisions

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Abstract- AI-Fin Advisor is an artificial intelligence supported web application that helps a user to make smarter financial decisions by providing customized recommendations for investment strategies, budgeting, debt management, and long-term wealth planning by analysing users income, expenditure, spending habits, savings, and financial goals. Natural Language processing, Machine Learning, and data analytics are used to provide insights and further decision-making tailored to each users unique financial situations. The application also provides additional features like real time financial news updates, an interactive financial literacy module which enhances users knowledge, financial calculators, stock analyser and a conversational chat bot to provide immediate responses. The system is developed by using React-based frontend and a Flask backend, incorporating MongoDB, machine learning models, and external financial APIs.Unlike the traditional financial advisors which are costly or not customized to users preferences, this solution provides real-time support that changes based on the financial circumstances. This research assists in offering financial advice that lends a hand for users to make data-driven financial advice despite their economic status or financial expertise.

KEYWORDS: Financial Literacy, Artificial Intelligence, Machine Learning, Natural Language Processing, Personalized Recommendations, Financial Planning, Robo- Advisory

I INTRODUCTION

Many individuals lack of access to financial advisor due to high costs, limited financial literacy, often resulting in poor money management and inadequate long-term financial stability. Furthermore, users financial data is frequently scattered across multiple applications and making it difficult to control savings, investments, and debt effectively. In today's increasing financial world, while many financial tools exist, most of them fail to provide individualized results, predictive insights, and integration with educational resources, leaving users without actionable metrics. There is a need for extensive, intelligent, and user favored financial advisory system.

This research presents AI-FinAdvisor, an AI-powered personalized financial advisor designed to aid users with data-driven financial insights which significantly enhance financial literacy. AI-FinAdvisor leverages natural language processing using Ollama, statistical models and machine learning algorithms to provide individualized recommendations to each financial profiles. By analyzing expenses, income, historical data, technical indicators, and predictive models, the platform AI-FinAdvisor, supports real-time stock trend forecasting and portfolio optimization , guiding informed investment decisions to user.

Beyond predictive analytics, AI-FinAdvisor also integrates financial educational content through video lectures, QNAs and a real-time RAG-based chatbot to interactively guide users. Privacy and Security aspects are ensured through AES encryption, SSL communication and bcrypt-based authentication, maintaining robust locked of sensitive financial data.

By combining secure data engineering, predictive analytics, AI and data engineering, AI-FinAdvisor bridges the gap between financial knowledge, practical decision-making, offering a scalable, intelligent resolution that makes personalized financial guidance accessible, actionable, and user-centric.

II RELATED WORKS

The use of AI in personal finance management systems has facilitated more personalized financial suggestions, projections, and engagement over time. Recent research

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provides evidence as to how AI modeling, machine learning, and natural language processing make financial decision-making processes, as well as advisory automation and customization, more streamlined.

A. Automated Advisory Systems & Financial Literacy

The research from [1] studies how personal finance transforming with machine learning , predictive analysis and natural language processing to assess income vs expenses vs savings for critical recommendations at all times. In the same vein, [2] presents Money Map, a personal finance tracking system for appropriate investments that champions smart expense tracking with impressive dashboards. It champions multilinguality with personalized finance based specific suggestions for inclusivity and comprehensive participation. The impact of financial literacy on financial well-being is studied in [3], emphasizing a mediating factor of personal finance management over how personally literate people will have better finances relative to people who are less literate. The authors of [4] expand on this with the GETU model-a policy consideration that champions financial literacy via government involvement, education efforts, training, and upskilling. [5] presents an AI-driven assistant for personal finance management for better financial literacy through natural language processes and understanding user feedback, while [6] emphasizes an AI-driven transformational personal finance advisory system for personalized-scale unique finance advice.

B. Machine Learning for Financial Forecasting and Decision Support

The study from [7] presents a flexible and adaptable hierarchical software architecture for machine learning in individual finance planning, introducing sub-structures for data collection, storage, learning and application visualization. Likewise, [8] champions how machine learning can support corporate financial forecasting and analysis while also criticizing machine learning in resource allocation for improvements during the capital budgeting process; Double Machine Learning(DML) is recommended as a framework to bolster substance to prior findings.

In [9], the authors integrates sentiment analysis of financial news and social media with conformist ML models for market trend prediction, achieving 98.9% accuracy using 98.4% using SVM and logistic regression . Another study [10] tells about usage of Facebook's Prophet model to forecast stock prices, allowing investors to make more well-versed and accurate trading verdicts.

The authors in [11] design an AI-based Decision Support System (DSS) for optimizing personal financial management. It integrates AI algorithms in the workflows for decision making, which significantly improves budgeting, expenditure control and investment while maintaining data privacy.

C. LLM and RAG-Based Conversational Financial Systems

Recent improvements in integrate Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) to enhance intelligent financial conversation. Paper [12] introduces QBot, a domain-specific chatbot uses vector embeddings and RAG pipelines to provide precise, context-aware replies. Correspondingly, [13] presents a Generative AI-powered financial advisory system combined with chatbot boundaries to deliver individualized and engaging advisory experiences.

The research in [14] inherits RAG capabilities by combining multimodal data (video,text, document) through LLaMA3 framework, enlightening interaction flexibility and understanding. Further, [15] smears RAG-LLMs to enhancing adaptability, financial risk management, and reliability by incorporating instantaneous market and sentimentality data.

Paper [16] advances a robo-advisory framework combining inverse optimization with deep understanding of reinforcement learning, allows AI to optimize portfolios by balancing risks vigorously. Study [17] explores trust dynamics in AI-driven retail financial investment services, which emphasizes the reputation of transparency and user assurance in automated advisory platforms.

D. Ethical, Privacy, and Societal Implications in AI Finance

Ethical and data confidentiality issues in economic technology are spoken in [18], which highlights fairness, transparency, and accountability via explainable AI and bias-mitigation approaches. The highlights in [19] enhances this by offering intelligent algorithms for financial data privacy fortification and risk prediction using federated learning, allowing collaborative model learning among banks without data leak.

In [20], the authors analyze how financial literacy influences college student's digital risk aversion and online financial security behaviors. Their conclusions reveals that higher financial awareness associates absolutely with alert online decision-making and security aware behavior.

Collectively, these studies highlights, that collaborating AI and ML into finance weights ethical, secure, and transparent frameworks that rank user trust, knowledge, and financial inclusion.

III SYSTEM OVERVIEW AND ARCHITECTURE

The proposed Ai-Fin Advisor follows three layer architecture consisting of frontend, backend, data intelligent layer. The frontend is where users input their financial data, track progress and receive personalized insights in an interactive interface. The backend is built with flask, which handles data management and facilitates communication between frontend, database and external API. The data intelligence layer is where all analysis and computations are done using AI models for financial advice, MongoDB for secure data storage and tools like recommendation module and stock analyzer to deliver insights.

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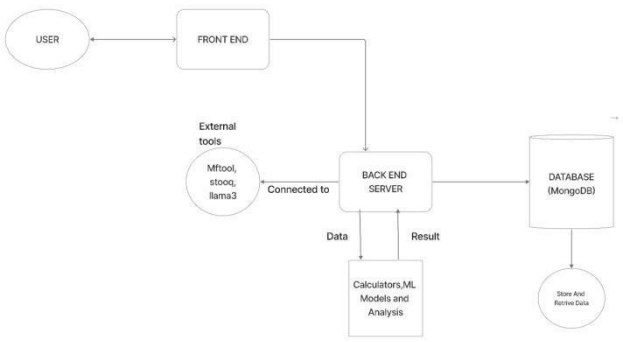


Fig. 1 Architecture of AI-Fin Advisor

As displayed in the fig. 1, we can see that the system architecture comprises of different modules as the following.

A. User Data Input Module

Users will provide relevant information about their finances including their assets(cash, real estate, gold, bonds), debts(credit card, student loans, personal loans), goals(short term and long term) and transactions (income, expenses, investments). The users are given an option to upload their monthly transactions file instead of manually entering the transactions. This data will be used to generate personalised recommendations and inform financial decisions. It uses an unstructured database, MongoDB, in order to store the users data.

B. Data Security

Since Ai-FinAdvisor deals with users sensitive data, it is very crucial that the system maintains data integrity and security. Multi factor authentication is achieved through email-based One-Time Passwords (OTP), with the SMTP protocol that facilitates automated OTP delivery. In order to securely save users credentials Bcrypt method is used for password hashing which adds unique salts to each password. The users data is securely stored by using the Fernet function in python's cryptography library which allows encryption during storage and decryption when required.

C. External Tools and Libraries

The proposed system integrates several external tools and libraries to enhance its functionalities. In the Stock Analyzer module, it utilizes Stooq API to fetch real time stock data while MFtool API is used to retrieve mutual fund information. The money matters provides the user with up-to-date financial related news and market updates by using the NewsAPI application. The financial literacy module incorporates resources from YouTube videos or publicly available documents.

IV METHODOLOGY

A. AI Chatbot

The Ai Chatbot system retrieves the data from MongoDb, where the users data is stored in an organised manner. It uses optimal queries to gather necessary data and feeds users historical data to the machine learning pipeline to find correlation and patterns. It interprets the query and generates text in human understandable language. It uses LLaMA3 model for the text generation. It also leverages NLP, the chatbot to forecast the future expenses or answer financial related queries and is also used in enhancing financial literacy.

B. Recommendations module

This module provides the financial suggestion system which incorporates both rule based and AI-driven insights to offer customized advice based on the user's financial data. It collects transactional, asset, debt and goals data of the users data by safely decrypting it and stores it in a structured dataframe and identifies spending patterns, category breakdowns using linear regression. In the rule based system it analyses debt and asset information. It produces reliable proposals such as giving importance to high interest debt payments, suggesting goal funding targets based on timelines.

C. Financial Literacy

This feature aims to educate the users about fundamentals of personal financial concepts along with the systems advisory features. The content is selected from trustworthy sources such as government portals, financial blogs, and YouTube videos. Topics are arranged according to the skill level (Beginner, Intermediate, Advanced) and stored in a database along with the metadata (title, tags, difficulty, quiz links). It also keeps track of the users learning progress. The application displays an interactive dashboard where user can browse topics, read or watch embedded videos, and take quizzes to reinforce learning. Chatbot is also there to assist by clarifying doubts.

V ALGORITHMS

A. Transactions File upload and categorization

Instead of manually entering the daily income and expenses regularly the system provides a feature of uploading transaction files in formats like CSV, TXT or Google Pay statements. The data is first cleaned by using regular expressions and customised functions. It is used to retrieve useful information like date, amount and description. Each transaction row is then automatically categorized using rule-based keyword matching and pattern recognition, assigning categories such as food, utilities, entertainment, transport, or savings. The users sensitive information such as amount, notes and date are encrypted using fernet before storage in MongoDB maintaining user privacy.

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B. **Rule-Based Recommendation Engine**

This algorithm relies on calculations or a set of pre-defined rules instead of machine learning. The calculator module computes loan EMIs, SIP maturity values, SWP withdrawals, or FIRE numbers using existing equations.

(i) EMI (Equated Monthly Installment):

$$EMI = \frac{P \times R \times (1+R)^N}{(1+R)^N - 1}$$

Where P = principal loan amount, R = monthly interest rate, and N = number of monthly installments.

(ii) SIP (Systematic Investment Plan) Maturity Value:

$$A = P \times \frac{(1+\frac{r}{n})^{nt}}{\frac{r}{n}} \times (1+\frac{r}{n})$$

Where A= Amount, P = Monthly investment, r = Annual rate of return, n = Compounding frequency (usually 12 months), t = Investment duration in years

C. Large Language Model Integration and Fine Tuning

The system integrates with Retrieval-Augmented Generation (RAG) framework with fine-tuning of a large language model to deliver high personalisation and grounded financial advice. The raw data is decrypted and fetched to the system. This data is enriched and then it is passed to a fine-tuned LLaMA-3 model which is used for generating text and has been adapted using anonymized profiles and expert-curated recommendations. By using the blend of both the system is able to handle analytical questions like “When can I retire?”, or “Predict my next month’s expenses.”.

D. Machine learning Algorithms for Stock Analyser

The stock analyser module includes the price prediction of stock and mutual funds that takes systematic data-driven modeling using multiple machine learning algorithms. The historical data is collected from Stooq (for stocks) and MFAPI (for mutual funds). The raw data is preprocessed which includes cleaning, handling missing values, and feature scaling using MinMaxScaler. Feature engineering generates indicators such as Simple Moving Average (SMA), Exponential Moving Average (EMA), Moving Average Convergence Divergence (MACD), Relative Strength Index (RSI), Bollinger Bands, lagged features, and volatility measures.Various machine learning algorithms are implemented that includes Linear Regression, Random Forest, Gradient boosting regressor and enhanced XGBoot. These models are trained by 80% of the historic data and the remaining 20 % is used for testing. A weighted ensemble model is constructed by combining predictions from the top three models based on their R^2 scores to improve generalization. Hyperparameters for each model are optimized through cross-validation to ensure stable and accurate price forecasting.

VI RESULTS ANALYSIS

Most of the financial advisory systems often rely on users to input data manually, have limited degrees of personalization, and lack AI features and insight generation. Prior work has tended to be either focused on expense tracking or financial planning and does not provide a unified intelligent solution. This work and project addresses the gap by providing an avenue for the automated upload of transactions, thus reducing the time needed to enter transactions manually by at least 85%, while also offering AI categorizations at approximately 90% accuracy, which is far superior to basic rule gaming categorization. To provide insight related to user predictions, the predictive modules incorporated in the system demonstrate good overall correlational trends forecast (R^2 =0.83, MAPE=17%) allowing for better user financial projections and decisions regarding investment and saving. More accurate, higher-quality forecasting modules then lead to improved user decision making and participation through interactive dashboards, effective goals based planning, and real-time recommendations in providing engagement, awareness, and efficiency. All together the predictive modules demonstrated summary improvements in terms of accuracy, user decisions and financial decision making providing some meaningful diagnostics of gaps demonstrated by previous work.

VII OUTCOMES

A. Stock Analyser Module

It asks the user to input the preferred stock ticker name and the time duration to analyse the performance history of the stock.

Model	R^2 Score	RMSE	Accuracy	MAPE
Ensemble (Top 3)	-4.3618	0.3171	76.17%	23.83%
Gradient Boosting	-1.7467	0.227	83.03%	16.97%
Linear Regression	1	0	100.00%	0.00%
Random Forest	-1.8905	0.2329	82.42%	17.58%
XGBoost (Enhanced)	-1.9751	0.2362	82.34%	17.66%

Table 1 Model performance of various algorithms

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Fig. 2 Output of Netflix stock price prediction

The stock price prediction of Netflix for 30 days is displayed in Fig.2. In order to increase the prediction accuracy the model applied, consists of historical price data, Moving Averages and technical indicators, such as Relative Strength Index (RSI). The models that were tested included; Linear Regression, Random Forest, Gradient Boosting, XGBoost, and an ensemble of the top three models. From the prediction in Table 1, Gradient Boosting yielded the highest predictive accuracy of 83.03% with a low MAPE (Mean Absolute Percentage Error) of 16.97%. The Ensemble approach also demonstrated comparable performance when weighing our results against the models individually. Linear Regression showed 100% accuracy on training data but appears to overfit the training data and may fail to generalize in the future. Random Forest and XGBoost also met the prediction accuracy criteria, yielding over 82% accuracy. Overall these results indicate that forecasting using a machine learning approach is effective in attempting to predict stock price trends over a short period, and therefore would be useful in regards to any investment or portfolio management considerations.

B. Financial Literacy Module

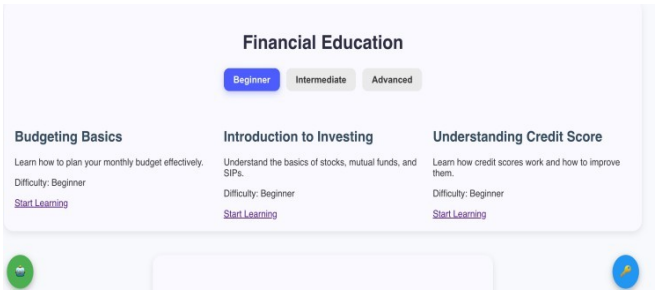


Fig. 3 Financial literacy demonstration

The Fig. 3 shows a overview of the financial literacy module in the AiFin Advisor application that presents the users with structured and interactive modules to improve their financial knowledge. The interface ensures adapting learning by classifying educational content into Beginner, Intermediate, and Advanced levels. Each content includes concise educational material either in video or document form and a quiz-based evaluation system. The users progress is also monitored and stored in the database. A real-time AI powered Chatbot is also integrated to the system which enables the user to ask financial related queries and receive immediate response.

VIII CONCLUSION AND FUTURE WORK

The AI-FinAdvisor introduces an intelligent approach to enhance users for managing and optimizing their personal finances with integration of ML algorithms, statistical methods and domain specific natural language processing through ollama for adapted recommendations. It provides prediction and portfolio optimization by analyzing user financial data and also provides users to improve their financial literacy through video lectures and quizzes. It screens technical pointers for real time market data to provide further insights. It guarantees data security by using AES for encrypting data and uses bcrypt method for secure authentication and SSL for communication. This study shows the potential of combining AI, finance and secure data engineering to help users in making smarter financial decisions. Future improvements could include integrating the system with secure banking APIs that will ensure automatic syncing of user's financial data without manual data entry. The user assignation and accessibility can be improved by incorporating voice-based interactions. Blockchain technology can be involved which will provide vigorous data safety and veracity. An AI-FinAdvisor modified financial advisor that integrates machine learning, natural language processing, and interactive web technologies to improve financial literacy and decision-making.

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